Project 02

PneuAssist

Wearable Device / Human-Computer Interaction / Assistive Technology

BRIEF

The main goal of this project is to **explore the use of soft pneumatic materials in rehabilitation**. It monitors **muscle activity** via EMG and provides gentle corrective force to **relieve hand fatigue** from poor working posture, while ensuring comfort for long-term use and supporting functional training and ergonomic protection.

INDIVIDUAL PROJECT

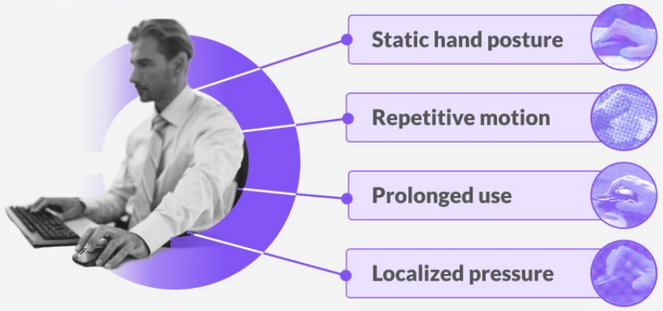
DURATION

sep. 2025 – Jan. 2026

BACKGROUND

With the rise of the information age, prolonged use of **computers, mice, and trackpads** has become standard in learning and work.

■ Hand Use



Most activities **maintaining the same hand posture** and relying on **frequent, precise hand input** for extended periods.

■ Hand Fatigue



1 Tenosynovitis & carpal tunnel syndrome

impact comfort and long-term health, yet remain difficult to fully resolve.

Their persistent yet 'not severe' nature causes them to be underestimated,

Resulting in **prolonged hand sub-health.** **2**

RESEARCH

■ Current Situation

*Data from ***Health Science Journal***

CTS (Carpal tunnel syndrome) prevalence



≈ 14.4 %

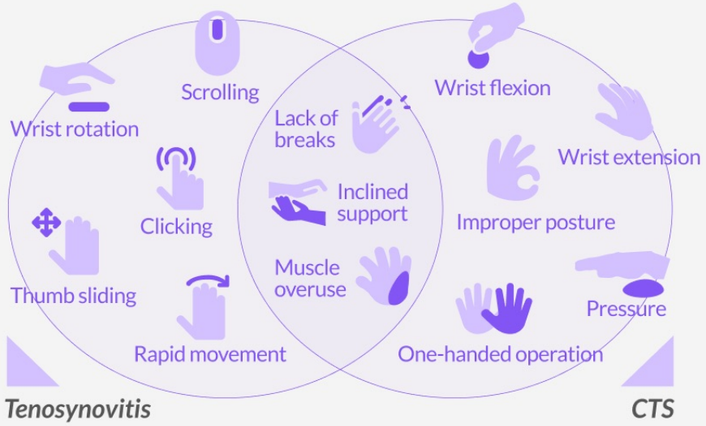
Among **508** participants tested positive for tenosynovitis **36.8%**

Tenosynovitis cases in the **dominant hand** was as high as **95.6%**

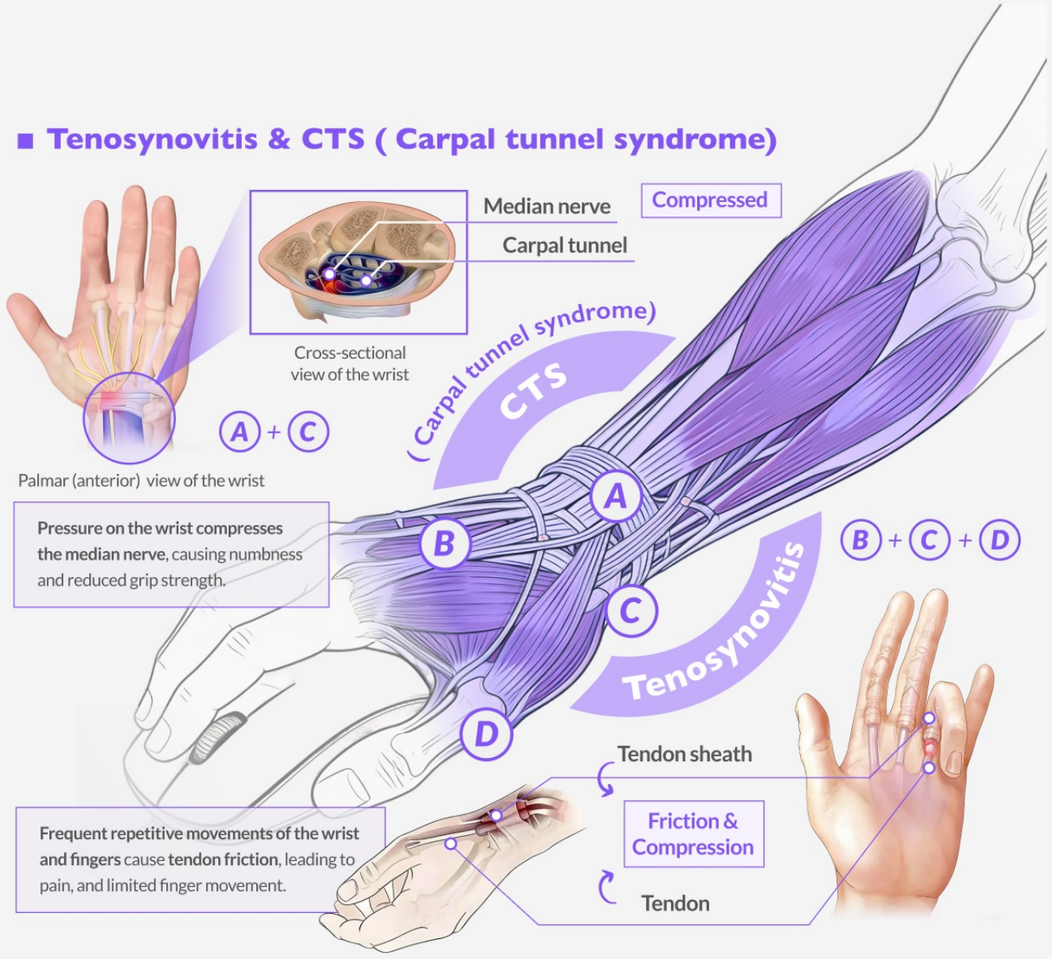
OR = Odds in exposed/unexposed

Exposure Factor	OR
Mouse use ≥20 h/week	1.93
Continuous work	1.89
Digital device use ≥6–8 h/day	1.34
Multiple years of computer work	1.92

■ Causes Behavior



■ Tenosynovitis & CTS (Carpal tunnel syndrome)



These prompted me to reconsider hand usage & input interactions, and to explore new ways to reduce hand and wrist strain.

IDEATION

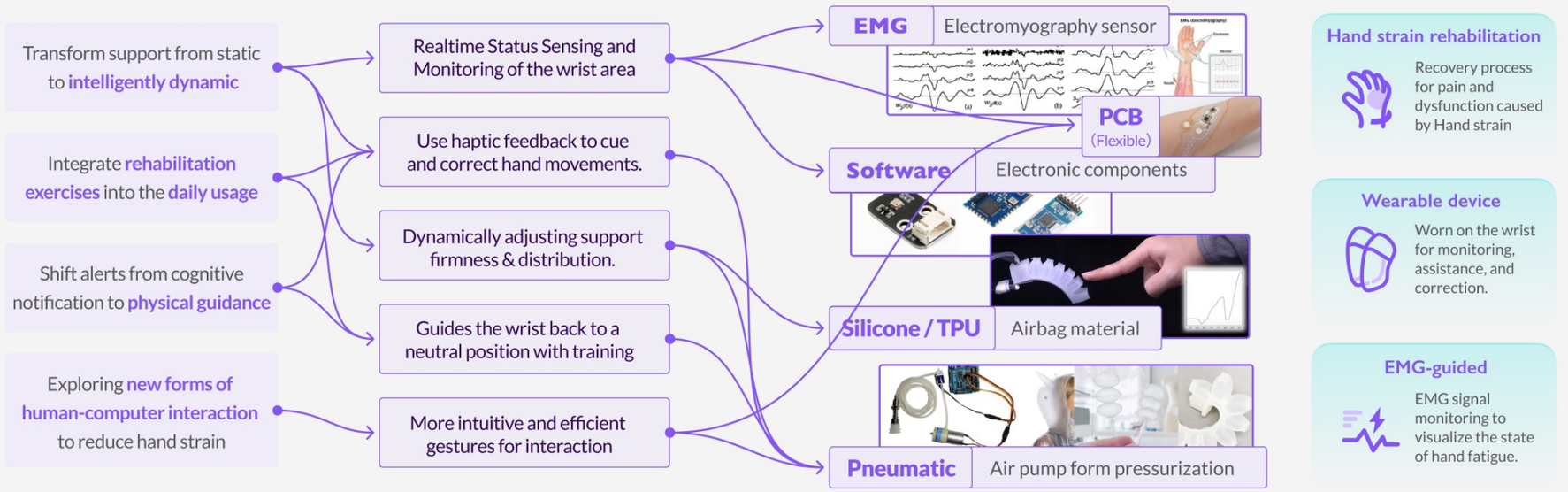
■ Insights

Cumulative Damage
The damage is done silently by micro-stress, long before pain signals the problem.

Postural Imbalance
Improper posture is unconsciously sustained during prolonged interaction.

"Efficiency" and "Health"
Users often sacrifice long-term health for immediate productivity or comfort.

Ineffective reminders
While many know they should take breaks, but conventional reminders are easily ignored.



TECHNICAL REASERCH

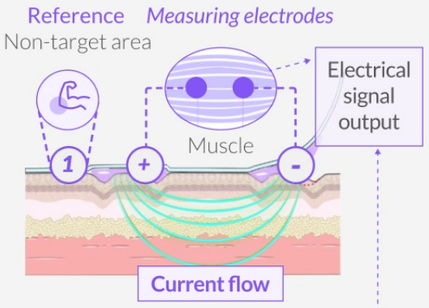
[1] Junwei Li et al.Flexible multichannel muscle impedance sensors for collaborative human-machine interfaces.Sci. Adv.11,eadv3359(2025).DOI:10.1126/sciadv.adv3359

How can we **sense hand strain earlier than pain, intervene gently, and retrain hand behavior**?

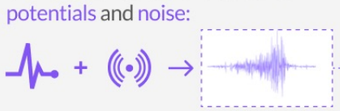
■ sEMG (Serface Electromyographic)

Captures electrical signals generated by muscle activation through skin-mounted electrodes, reflecting muscle activity patterns.

01 Operating principle

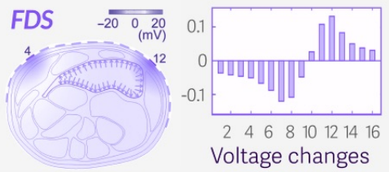


sEMG is composed of both action potentials and noise:

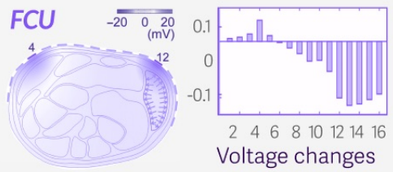


02 Measurement data

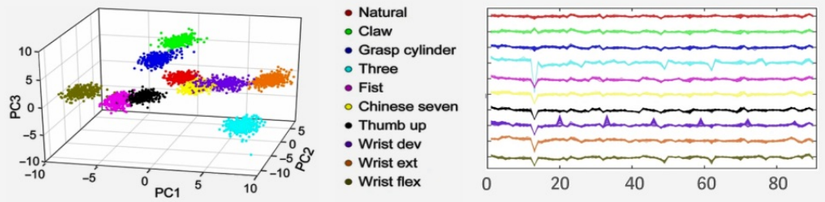
Fexor digitorum superficialis controls **finger flexion.**



Flexor carpi ulnaris controls **wrist movement.**



sEMG Signal Waveforms Under **Different Hand Activities**



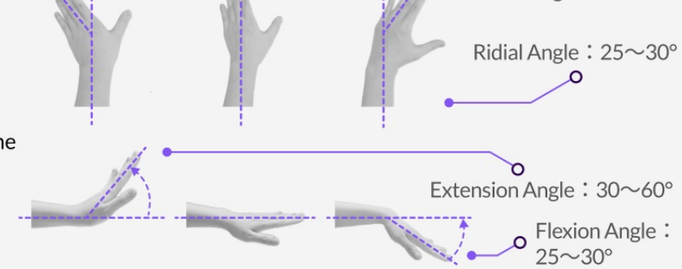
■ Ergonomic

01 Hand gestures

Different types of mouse grip postures



02 Wrist rotation range

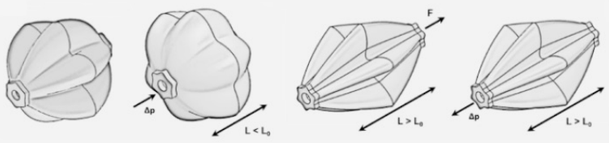


Based on the analysis of hand, I focus on using gentle guiding forces to **correct long-term deviations from a neutral hand posture.**

■ Pneumatic

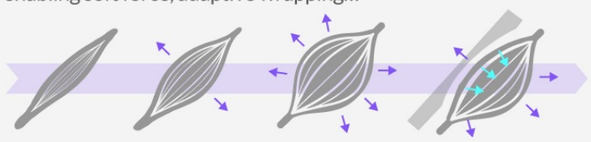
Uses pressurized air to create motion, force, or shape changes in a flexible and controllable manner.

01 Basic principle



02 Operating

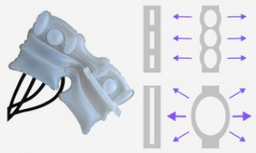
Complex deformations such as bending, extension can be achieved, enabling soft force, adaptive wrapping…



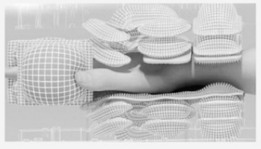
03 Morphology Exploration



Different material properties (softness, thickness) influence force direction after inflation



The shape and size also affect the magnitude and direction of force.



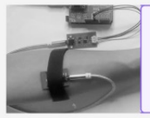
By designing the air bladder's shape, size, and material, combined with selective inflation, **gentle forces can be applied in multiple directions.**

ERGONOMIC STUDY

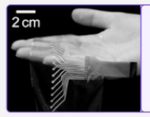
■ Feasible Testing



Wet Electrode
High accuracy & Short-term use



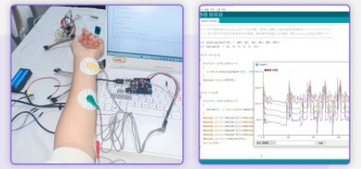
Rigid Dry Electrode
Long lifespan & Body mismatch



Flexible Dry Electrode
High comfort & Higher cost

FDS

Measuring signals during **finger flexion/ extension** and **wrist movement.**

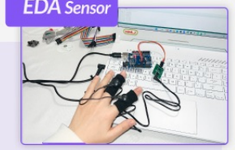


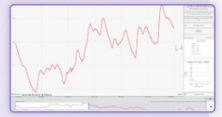
APL

Measuring muscle activity **in the thumb and its first joint.**



EDA Sensor





Electrodermal activity sensors were also tested but showed limited ability to reflect muscle fatigue.

■ Participatory Study Workshop

A "body-centred" approach to user research

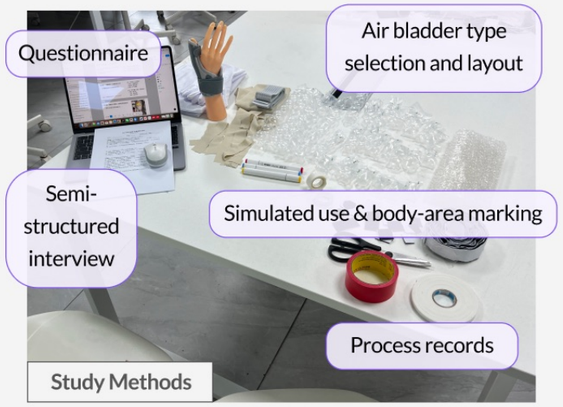
Since this design involves wearable technology, ergonomics and personalized wearing comfort are critical considerations. Therefore, I conducted workshops to better capture users' experiential needs.



Materials & Tools



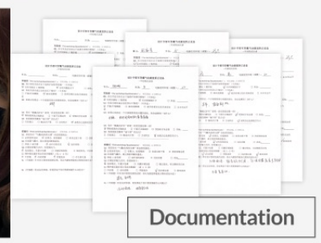
Workshop Process



Study Methods







Documentation

■ Design considerations

Flexible and Compliant
Wearable devices need to remain flexible and compliant to support fine hand movements, instead of restricting the hand like rigid exoskeletons.

Lightweight
The wearable components should prioritize lightweight design to avoid unnecessary burden on movements.

Gentle Force
It should be easy and intuitive to wear, minimizing wrapping and restriction on the hand, and allowing the user to complete the process single-handedly.

Easy to Wear
It should be easy to put on, minimizing wrapping and restriction on the hand, and allowing it to be worn with just one hand.

EXPLOREATION

By analyzing different types of workshop results, we can identify the **specific areas where users need** support in this product.

■ Workshop Analysis

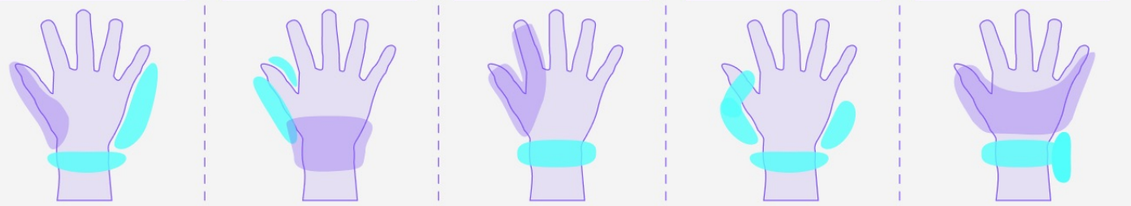
Large thick airbag · Small thin airbag · Lack of support · Prone to fatigue

Workshop Results



Thumb movement, pinky-side support | Thenar tension & unsupported wrist | Repetitive thumb & index movement | Overall unstable palm structure | Dynamic instability frequent motion

Distribution



Area Statistics



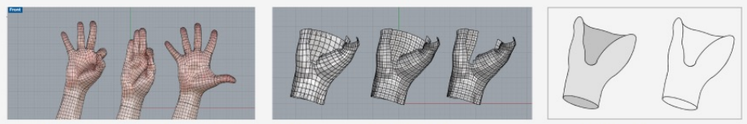
Unfolded Surface



■ Model process

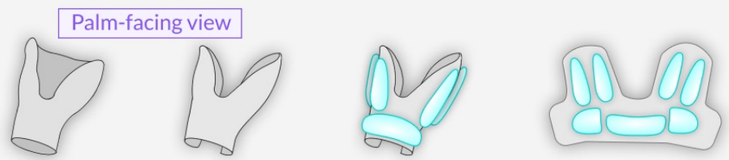


Fabric models test



3D scan hand mesh | Primary distribution area | Wireframe

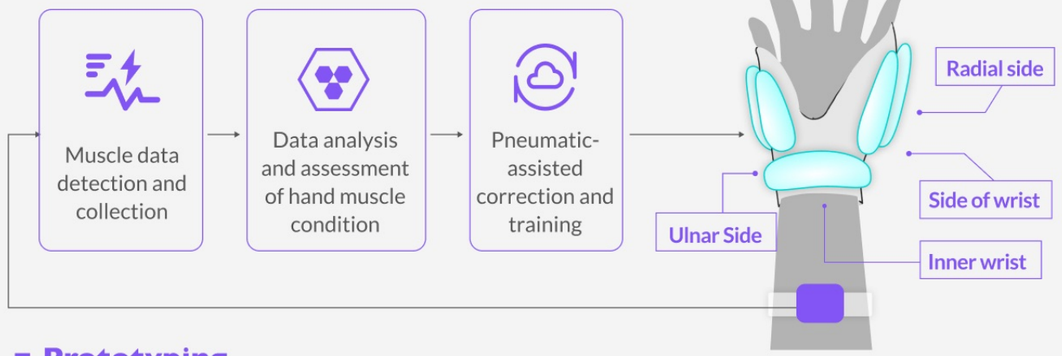
Palm-facing view



Simplified | Airbag distribution | Curve tracing

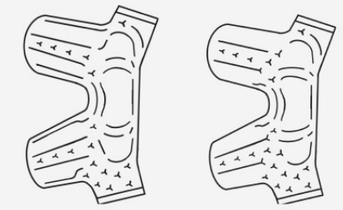
By analyzing the workshop results and testing fabric models, I **build a basic shape and distribution mode** of this wearable device.

■ Architecture



■ Prototyping

01 Curve tracing



02 Pattern transfer and cutting



03 TPU heat bonding



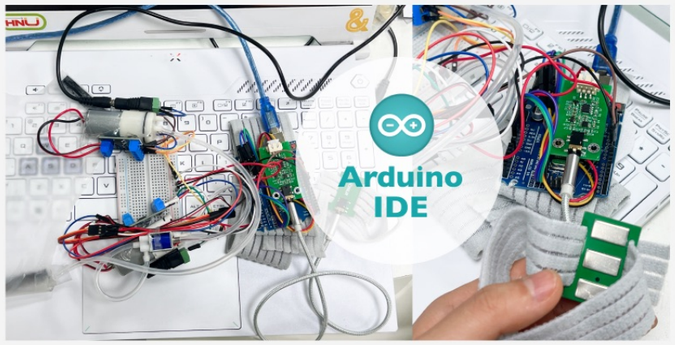
04 Forming and inflation test



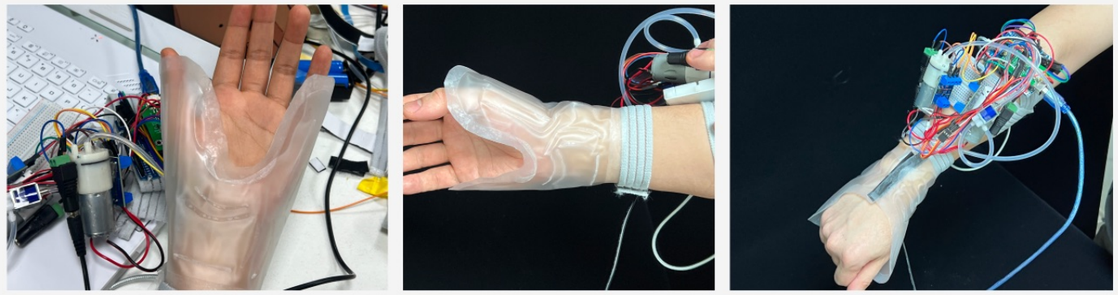
05 Coding

Using Arduino to connect sensors and pump to create an interactive **hardware prototype.**



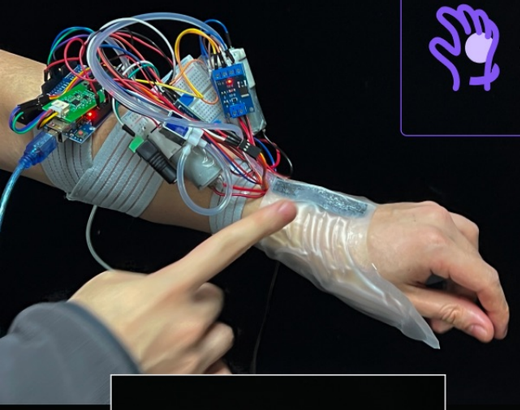


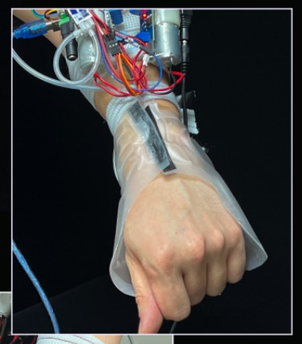
■ Testing

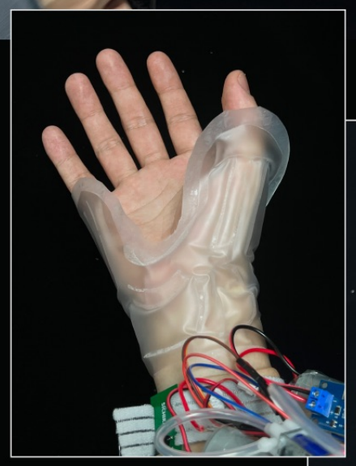


OUTCOME

The final design is a **wearable device** wrapping the wrist and sides of the hand, offering gentle correction for **fatigue-induced improper hand postures.**

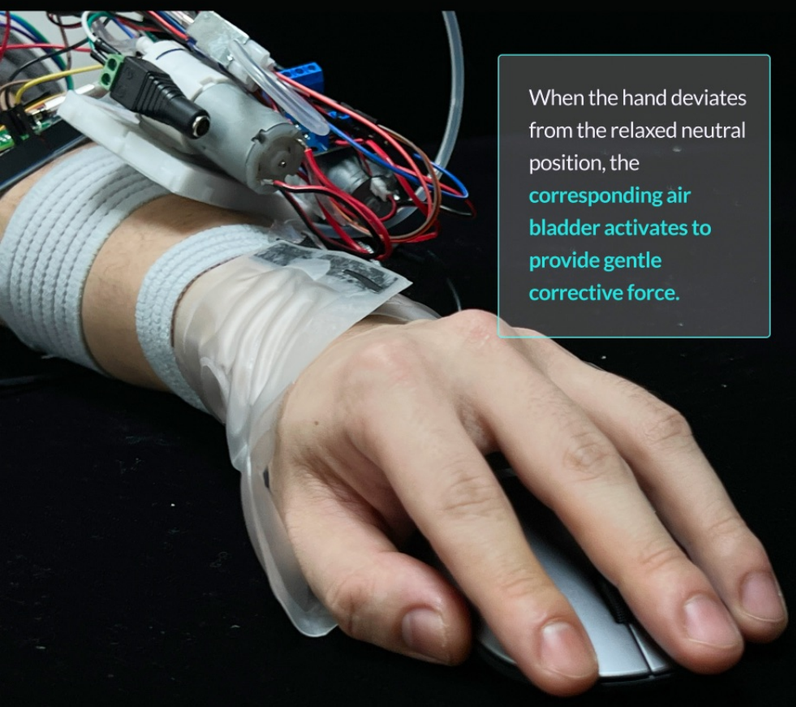






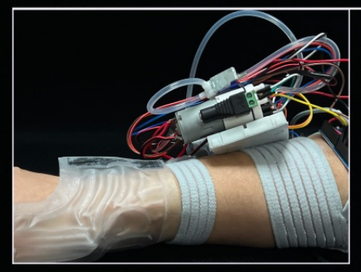


■ Usage



When the hand deviates from the relaxed neutral position, the **corresponding air bladder activates to provide gentle corrective force.**

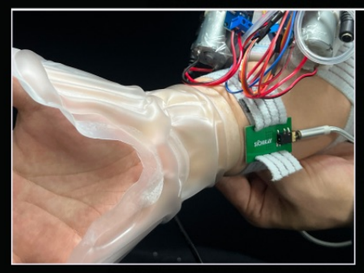
• Pneumatic Device



The core control unit of the overall system

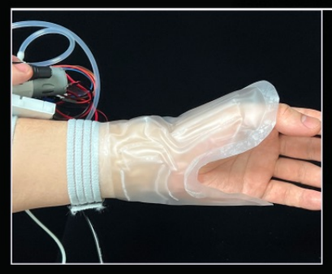
Generates controllable air pressure for physical actuation and support.

• EMG Detection

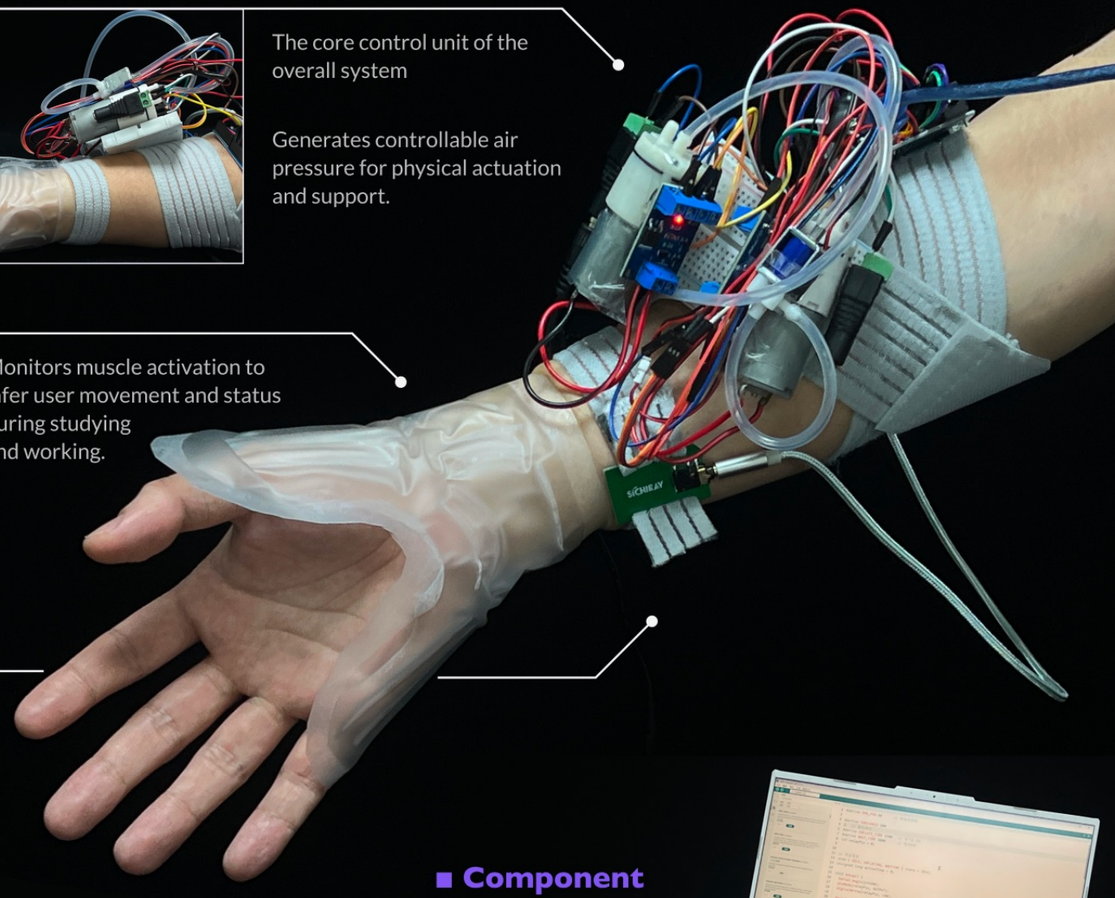


Monitors muscle activation to infer user movement and status during studying and working.

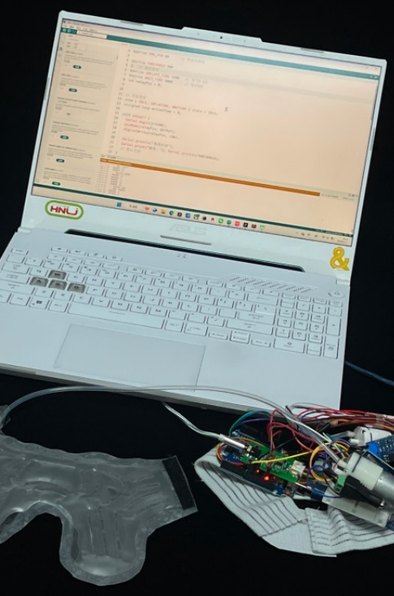
• Wearable Corrective Device



A soft wearable device for behavior guidance and correction

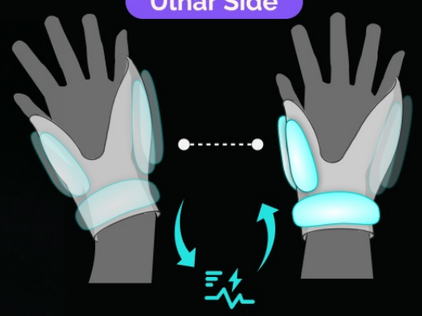


■ Component

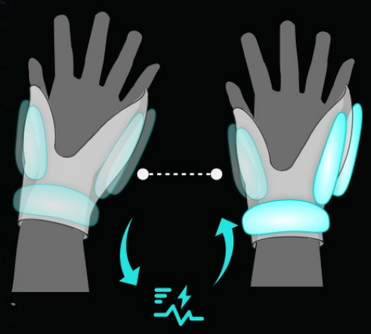


- **Arduino board**
  Acts as the system brain, processing signals and controlling components.
- **Relay Module**
  Uses a low-power signal to switch high-power devices on and off.
- **Micro Air Pump Module**
  Provides air pressure for inflation and actuation.
- **Dry Electrode EMG Sensor**
  Detects muscle electrical signals without conductive gel.
- **Solenoid Valve**
  Precisely controls airflow direction and release.

Ulnar Side



Radial side



REFLECTION

This approach appears highly feasible, but implementing it in practice still faces challenges.

Pneumatic materials are non-breathable, making prolonged wear uncomfortably hot, and the air pump is too large and heavy for wearable devices, so a more compact and convenient alternative is needed.